💻 Project Story: Laptop Price Predictor

📌 Objective

The objective of this project is to build a **Machine Learning Model of Regression** that can predict the price of a laptop based on its hardware specifications. With the massive variety of laptops in the market, choosing the right one at the right price can be confusing. This project aims to simplify that decision by providing an estimated price based on key configurations like processor speed, RAM, and storage.

🧠 Problem Statement

Laptop prices are influenced by many hardware and brand-related factors. For a typical buyer, it's not always easy to evaluate if a specific configuration is priced fairly. This project addresses that problem by using machine learning to predict the approximate market price of a laptop based on its technical specifications.

📦 Dataset Details

- **Name**: Laptop Price Dataset
- **Source**: Public dataset or custom data collection
- **Attributes Used**:
    - Processor Speed (GHz)
    - RAM Size (GB)
    - Storage Capacity (GB)
    - *(More features can be added like brand, GPU, screen size, etc.)*
- **Target Variable**: Price

The dataset includes laptops from various brands with different hardware combinations to ensure a diverse and representative training base.

🛠️ Project Workflow

🔍 1. Data Collection & Preprocessing

- Collected or used an existing dataset containing laptop specs and their prices.
- Cleaned the dataset, handled missing values, and standardized numerical data.

- Selected the most impactful features for predicting laptop price.

⚙️ 2. Model Development

- The problem was treated as a **regression task**.
- Trained multiple models, and selected the **Random Forest Regressor** due to its accuracy and robustness.
- Used train_test_split to divide data and validate performance.

📈 3. Model Evaluation

- Evaluated the model using:
    - Mean Absolute Error (MAE)
    - Mean Squared Error (MSE)
    - Root Mean Squared Error (RMSE)
    - R^2 Score
- Achieved high predictive performance and consistent results.

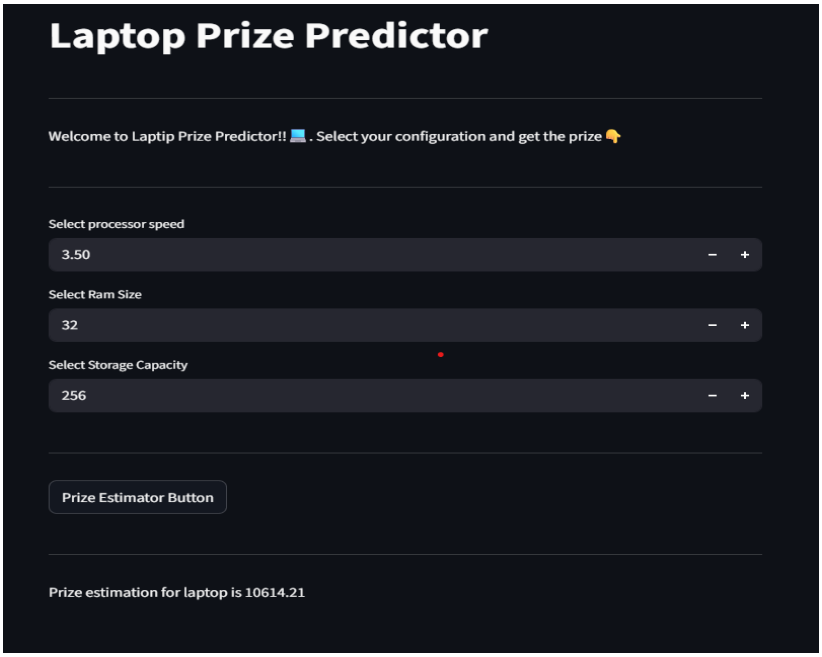
🌐 4. User Interface with Streamlit

- Deployed the model using **Streamlit** for a clean, interactive web interface.
- Users can input:
    - Processor Speed (GHz)
    - RAM Size (GB)
    - Storage Capacity (GB)
- On clicking the **"Prize Estimator Button"**, the model instantly returns a predicted price based on the given configuration.

📸 Final App Snapshot

The interface, as shown in the final screenshot, reflects:

- Clean, modern UI with dark theme
- Easy-to-use sliders for input
- Instant prediction output



🧰 Tech Stack Used

- **Language**: Python
- **Libraries**:
  - Pandas, NumPy – Data preprocessing
  - Scikit-learn – Machine learning
  - Streamlit – Web deployment
  - Matplotlib/Seaborn – Data visualization (for EDA)

✅ Project Outcomes

- Built a working, ML-powered **Laptop Price Estimator**
- Successfully deployed the model with an easy-to-use interface
- Gained deep experience in:
  - Regression modeling
  - Feature selection
  - Streamlit UI development

🔮 Future Scope

- Add more features: Brand, GPU, Screen Size, Touchscreen
- Deploy online using platforms like Heroku or AWS
- Extend the app to mobile-friendly versions